

CALVIN MICKELSON

SOFTWARE DEVELOPER

SOFTWARE
DEVELOPER
PROJECT RESUME

EMAIL mickelsoncalvin021@gmail.com

PHONE 970-666-1069

LOCATION Fort Collins, CO

GITHUB cabbabbage

LINKEDIN Calvin Mickelson

PORTFOLIO Portfolio site

SUMMARY

Fort Collins-based software developer with a B.S. in Computer Science and hands-on experience building performance-sensitive C++ systems, custom rendering pipelines, developer tooling, real-time applications, and machine learning/computer vision projects. Strong in C++, Python, and JavaScript, with a focus on fast, debuggable software that works under real constraints.

TECHNICAL SKILLS

LANGUAGES

C++, Python, JavaScript, TypeScript, Java, SQL, C, HTML/CSS, Bash

FRAMEWORKS/TOOLS

React, Node.js, Git, CMake, TensorFlow, NumPy, Pandas, SDL, MediaPipe

CORE STRENGTHS

Software engineering, data structures and algorithms, systems design, performance optimization, debugging, developer tooling, rendering pipelines, procedural generation, asset tooling, machine learning

SELECTED PROJECTS

ENGINE 2.0 / VIBBLE Engine

C++, SDL, CMake, JSON

- Built a custom 2D engine with dynamic lighting, parallax rendering, layered world composition, and clear separation between runtime, editor, and tooling systems.
- Implemented procedural map generation with room layering, sector-based placement, parent-child room relationships, collision/interaction geometry, and JSON-driven assets.
- Developed editor tooling for asset management, rendering controls, animation caching, and scalable content pipelines supporting rapid iteration.

auto_edit

Python, OpenCV, librosa, FFmpeg, EasyOCR

- Built an automated video generator that scans audio/media inputs, detects beats, drop timing, and bass energy, then schedules edits against the music timeline.
- Added OCR-driven text overlay preprocessing, configurable glitch/transition effects, dry-run timeline debugging, and batch rendering of numbered vertical MP4 variants.

ASL Interpretation Model

Python, TensorFlow, MediaPipe

- Built a real-time ASL recognition pipeline using hand landmark extraction and classification from live video input.
- Trained and evaluated models for sign prediction and implemented preprocessing to improve inference stability and usability.

SSH_MAVLINK

Python, Raspberry Pi, MAVLink

- Developed a remote multirotor control system over 4G LTE with real-time telemetry, video streaming, and command/control links.
- Enabled remote manual and autonomous flight workflows using MAVLink-based communication.

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SELECTED PROJECT

SmileScope

Python, MediaPipe

- Built a real-time facial landmark application to detect smiles and capture annotated screenshots under noisy motion conditions.
- Used MediaPipe landmarks and Python image processing to support lightweight computer vision interaction experiments.

EXPERIENCE

Broadcom - Manufacturing Control

Jan 2026 - Mar 2026

Fort Collins, CO

- Supported manufacturing control operations in a high-volume semiconductor production environment.
- Worked across schedule-driven workflows and internal production systems to help maintain continuity and process visibility.

Collins Control and Electric - Electrician Apprentice

2022 - 2023

Fort Collins, CO

- Assisted with electrical and generator installations and supported troubleshooting of networking, firmware, and control-system issues.

UNION Bar & Soda Fountain - Line Cook / Prep Cook

2022 - 2025

Fort Collins, CO

- Worked in a fast-paced, high-volume environment while coordinating prep, line execution, and quality standards across daily service.

EDUCATION

Colorado State University - B.S. Computer Science

Graduated 2025

Fort Collins, CO

Coursework and project work centered on software engineering, algorithms, systems, data, and applied machine learning.